Docker & Docker Compose Cheat Sheet

Java Microservices Project

Stack Overview

Service	Container
API Gateway	java-api-gateway-1
Employee Service	java-employee-service-1
Cache Service	java-cache-service-1
PostgreSQL	java-sql-db-1
Redis	java-redis-db-1
Frontend UI	java-frontend-ui-1

Docker Commands (Container-Level Operations)

View Containers

```
docker ps
```

Show all running containers.

```
docker ps -a
```

Show all containers including stopped containers.

View Logs

```
docker logs java-employee-service-1
```

```
docker logs -f java-employee-service-1
```

```
docker logs -f java-cache-service-1
```

```
docker logs -f java-api-gateway-1
```

```
docker logs -f java-sql-db-1
```

```
docker logs -f java-redis-db-1
```

Enter Containers

Employee Service

```
docker exec -it java-employee-service-1 sh
```

Cache Service

```
docker exec -it java-cache-service-1 sh
```

Postgres

```
docker exec -it java-sql-db-1 sh
```

Redis

```
docker exec -it java-redis-db-1 sh
```

API Gateway

```
docker exec -it java-api-gateway-1 sh
```

Frontend

```
docker exec -it java-frontend-ui-1 sh
```

Start / Stop / Restart

```
docker stop java-employee-service-1
```

```
docker start java-employee-service-1
```

```
docker restart java-employee-service-1
```

Inspect Container

```
docker inspect java-employee-service-1
```

```
docker inspect java-sql-db-1
```

Monitor Resources

```
docker stats
```

Single container:

```
docker stats java-employee-service-1
```

Execute Commands

```
docker exec java-employee-service-1 env
```

```
docker exec java-employee-service-1 ps -ef
```

```
docker exec java-employee-service-1 ls -l
```

Network Information

```
docker network ls
```

```
docker network inspect java_default
```

Volumes

```
docker volume ls
```

```
docker volume inspect <volume-name>
```

Cleanup

```
docker container prune
```

```
docker image prune
```

```
docker system prune
```

Docker Compose Commands (Application-Level Operations)

Start Entire Stack

```
docker compose up
```

Detached mode:

```
docker compose up -d
```

Rebuild everything:

```
docker compose up -d --build
```

Stop Entire Stack

```
docker compose stop
```

Remove containers:

```
docker compose down
```

Remove containers and volumes:

```
docker compose down -v
```

View Services

```
docker compose ps
```

View Logs

All services:

```
docker compose logs
```

Follow all logs:

```
docker compose logs -f
```

Employee Service:

```
docker compose logs -f employee-service
```

Cache Service:

```
docker compose logs -f cache-service
```

API Gateway:

```
docker compose logs -f api-gateway
```

Database:

```
docker compose logs -f sql-db
```

Redis:

```
docker compose logs -f redis-db
```

Enter Service Shell

```
docker compose exec employee-service sh
```

```
docker compose exec cache-service sh
```

```
docker compose exec sql-db sh
```

```
docker compose exec redis-db sh
```

Restart Services

Entire stack:

```
docker compose restart
```

Employee Service only:

```
docker compose restart employee-service
```

Cache Service only:

```
docker compose restart cache-service
```

Build Services

Build all:

```
docker compose build
```

Build Employee Service:

```
docker compose build employee-service
```

Build Cache Service:

```
docker compose build cache-service
```

Build without cache:

```
docker compose build --no-cache
```

Scaling

```
docker compose up -d --scale employee-service=3
```

PostgreSQL Commands

Connect to PostgreSQL

```
docker exec -it java-sql-db-1 psql -U postgres
```

List databases

```
\l
```

Connect database

```
\c employees
```

List tables

```
\dt
```

Redis Commands

Connect to Redis

```
docker exec -it java-redis-db-1 redis-cli
```

Ping Redis

```
PING
```

List keys

```
KEYS *
```

Get value

```
GET mykey
```

Daily Developer Workflow

Start stack

```
docker compose up -d
```

Verify

```
docker compose ps
```

Watch logs

```
docker compose logs -f
```

Debug service

```
docker compose exec employee-service sh
```

Restart service

```
docker compose restart employee-service
```

Check resources

```
docker stats
```

Stop stack

```
docker compose down
```

Full reset

```
docker compose down -v
docker compose up -d --build
```